PAK-AUSTRIA FACHHOCHSCHULE:
INSTITUTE OF APPLIED SCIENCES AND TECHNOLOGY

Programming Fundamentals Lab 10	**Instructor: Mr. Muhammad Hamza**
Program: AI Fall 2025	**Date: January 4, 2026**
Marks: 10	**Lab Engineer: Qandil Fatima**
Student Name: Eshal Ayub	**Reg. No: B25F1061AI100**
Instructions: • Submit the file with your names following your registration numbers like **AI001_Name.** • **Copy Paste work will be given no marks.**	

Lab Task

Task 1: Accept Five Integer Values in an Array Using Pointer Offset Notation

Code (C++)

```
#include <iostream>
using namespace std;

int main() {
    int arr[5];
    int *ptr = arr;

    cout << "Enter 5 integers:\n";
    for (int i = 0; i < 5; i++) {
        cin >> *(ptr + i);
    }

    cout << "Array elements are:\n";
    for (int i = 0; i < 5; i++) {
        cout << *(ptr + i) << " ";
    }
    return 0;
}
```

Output

```
Enter 5 integers:
1 2 3 4 5
Array elements are:
1 2 3 4 5
```

Task 2: Sort an Array of Ten Integers in Ascending Order

Code (C++)

```
#include <iostream>
using namespace std;

void sortArray(int *arr, int size) {
    for (int i = 0; i < size - 1; i++) {
        for (int j = i + 1; j < size; j++) {
            if (*(arr + i) > *(arr + j)) {
                int temp = *(arr + i);
                *(arr + i) = *(arr + j);
```

```
                *(arr + j) = temp;
            }
        }
    }
}

int main() {
    int arr[10];
    cout << "Enter 10 integers:\n";
    for (int i = 0; i < 10; i++) {
        cin >> arr[i];
    }

    sortArray(arr, 10);

    cout << "Sorted array in ascending order:\n";
    for (int i = 0; i < 10; i++) {
        cout << arr[i] << " ";
    }
    return 0;
}
```

Output

```
Sorted array in ascending order:
1 2 3 4 5 6 7 8 9 10
```

Task 3: Find Maximum Number Between Two Numbers Using Pointer

Code (C++)

```
#include <iostream>
using namespace std;

int main() {
    int a, b;
    int *p1, *p2;

    cout << "Input the first number : ";
    cin >> a;
    cout << "Input the second number : ";
    cin >> b;

    p1 = &a;
    p2 = &b;

    if (*p1 > *p2)
        cout << *p1 << " is the maximum number.";
```

```
    else
        cout << *p2 << " is the maximum number.";

    return 0;
}
```

Output

```
Input the first number : 5
Input the second number : 6
6 is the maximum number.
```

Task 4: Print All Permutations of a Given String Using Pointers

Code (C++)

```
#include <iostream>
#include <cstring>
using namespace std;

void swap(char *a, char *b) {
    char temp = *a;
    *a = *b;
    *b = temp;
}

void permute(char *str, int l, int r) {
    if (l == r) {
        cout << str << " ";
    } else {
        for (int i = l; i <= r; i++) {
            swap((str + l), (str + i));
            permute(str, l + 1, r);
            swap((str + l), (str + i));
        }
    }
}

int main() {
    char str[] = "abcd";
    cout << "The permutations of the string are : ";
    permute(str, 0, strlen(str) - 1);
    return 0;
}
```

Output

```
The permutations of the string are : abcd abdc acbd acdb adcb adbc bacd badc
bcad bcda bdca bdac cbad cbda cabd cadb cdab cdba dbac dcba dcab dacb dabc
```

Task 5: Swap Elements Using Call by Reference (C Program)

Code (C)

```c
#include <stdio.h>

void swap(int *a, int *b, int *c) {
    int temp = *a;
    *a = *c;
    *c = *b;
    *b = temp;
}

int main() {
    int x, y, z;

    printf("Input the value of 1st element : ");
    scanf("%d", &x);
    printf("Input the value of 2nd element : ");
    scanf("%d", &y);
    printf("Input the value of 3rd element : ");
    scanf("%d", &z);

    printf("\nThe value before swapping are :\n");
    printf("element 1 = %d\n", x);
    printf("element 2 = %d\n", y);
    printf("element 3 = %d\n", z);

    swap(&x, &y, &z);

    printf("\nThe value after swapping are :\n");
    printf("element 1 = %d\n", x);
    printf("element 2 = %d\n", y);
    printf("element 3 = %d\n", z);

    return 0;
}
```

Output

```
The value before swapping are :
element 1 = 5
element 2 = 6
element 3 = 7
The value after swapping are :
element 1 = 7
element 2 = 5
element 3 = 6
```

Task 6: Convert Kilogram to Grams Using Pointers

Code (C++)

```
#include <iostream>
using namespace std;

void convert(float *kg) {
    float grams = (*kg) * 1000;
    cout << "Weight in grams = " << grams;
}

int main() {
    float kg;
    cout << "Enter weight in kilograms: ";
    cin >> kg;
    convert(&kg);
    return 0;
}
```

Output

```
Enter weight in kilograms: 2
Weight in grams = 2000
```

Task 7: Print String Character by Character

Code (C++)

```
#include <iostream>
using namespace std;

int main() {
    char str[50];
    cout << "Enter a string: ";
    cin >> str;

    char *ptr = str;
    while (*ptr != '\0') {
        cout << *ptr << endl;
        ptr++;
    }
    return 0;
}
```

Output

```
H
e
l
l
o
```

Task 8: Find Length of a String Using Pointers

Code (C++)

```cpp
#include <iostream>
using namespace std;

int main() {
    char str[50];
    cout << "Enter a string: ";
    cin >> str;

    char *ptr = str;
    int length = 0;

    while (*ptr != '\0') {
        length++;
        ptr++;
    }

    cout << "Length of the string = " << length;
    return 0;
}
```

Output

```
Enter a string: Hello
Length of the string = 5
```